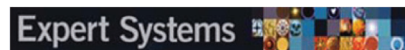


ORIGINAL ARTICLE



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Classifying the heart sound signals using textural-based features for an efficient decision support system

Kriti Taneja | Vinay Arora | Karun Verma

Department of Computer Science & Engineering, Thapar Institute of Engineering & Technology, Patiala, India

Correspondence

Kriti Taneja, Department of Computer Science & Engineering, Thapar Institute of Engineering & Technology, Patiala, Punjab, India.

Email: ktaneja20_phd17@thapar.edu

Abstract

Cardiovascular diseases have surpassed cancer as the leading cause of death on the planet today. Numerous decision-making systems with computer-assisted support have been developed to assist cardiologists to detect heart disease, and thus, lowering the mortality rate. The purpose of this research is to classify audio signals received from the heart as normal or abnormal. The PhysioNet Computing in Cardiology (CinC) 2016 benchmark dataset, popularly known as PhysioNet 2016, has been used to validate the proposed methodology presented here. PhysioNet 2016 contains a total of 3200 phonocardiogram (PCG) recordings divided into sub-datasets A-F. The state-of-the-art studies conducted till date have not considered the harmonic details of the beat that can be extracted from its equivalent chromagram image. In this work, textural features such as linear binary pattern (LBP), adaptive-LBP, and ring-LBP have been extracted from the existing spectrogram and combined with the features extracted from the chromagram. It has been observed that the combination of features extracted from both the image variants has resulted in a greater accuracy as compared to the scenario where researchers were using only the spectrogram. The experiment yielded the mean accuracy, precision, and F1-score as 94.87, 93.11, and 95.273, respectively. The heart sound classification models employ spectrogram, scalogram, and mel-spectrogram images to view and analyse the acoustic properties of a PCG signal. Although these visual tools provide useful information about the signal, yet they are unable to distinguish between pitch and resonance in heart sound generation. However, this paper proposes an alternative approach of heart sound signal representation that allows for a more precise measure of pitch-related changes in the heart sound. Its results highlight the significance of extracting textural features from time-chroma representation (i.e., chromagram) of PCG signals that have not been explored yet in the domain related to classification of heart sound signals as normal and abnormal.

KEYWORDS

adaptive local binary pattern, chromagram, decision support system, local binary pattern, PCG signals, ring local binary pattern, support vector machine

1 | INTRODUCTION

Machine learning is a significant subject in the healthcare domain. It addresses the real-world health problems by immersing itself in the analysis of medical data to forecast diseases. Machine learning techniques are currently being used to predict several diseases, including diabetes, cancer,

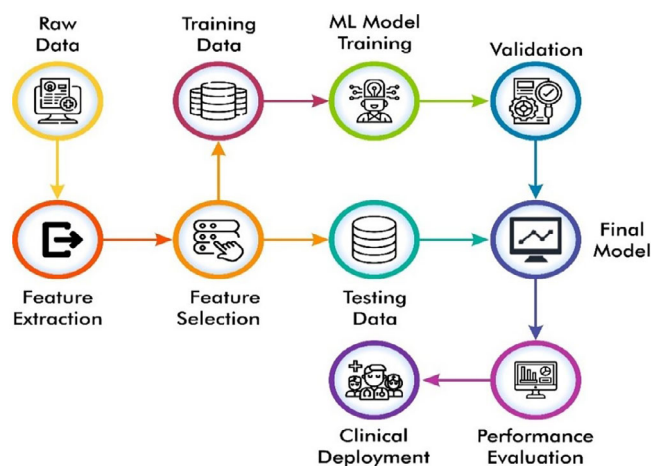


FIGURE 1 Machine learning approach in healthcare (Dwivedi et al., 2018).

stroke, and cardiovascular problems. Some machine learning algorithms including naive Bayes, decision tree, neural network, genetic algorithm, and support vector machine (SVM) have different accuracy levels in predicting these diseases (El Houbay, 2018). The steps involved in machine learning approach for predicting outcome in healthcare have been demonstrated in Figure 1. Some of the most dreadful and diverse diseases are caused by cancer, hepatitis, and cardiovascular problems. There is a huge amount of data that originates from instrumental and clinical research about these diseases. Thus, it is of utmost significance to develop effective tools which may assist in their management.

Cardiovascular disorders are now the biggest cause of death worldwide, surpassing even cancer and diabetes. In recent years, numerous methods have been examined in machine learning for cardiac health status detection. Electrocardiogram (ECG; Caesarendra et al., 2022; Celin & Vasanth, 2018), Photoplethysmography (PPG) (Ismail et al., 2022), and phonocardiogram (PCG; Dwivedi et al., 2018) are frequently used modalities in the monitoring of cardiac health. Given the convenience and feasibility of using a stethoscope to acquire heart rate data, PCG signals have been employed as raw data input to the classification model in the proposed work for detecting heart health status. In the literature, both supervised (Das et al., 2019; Moulana et al., 2022) and unsupervised (Das et al., 2020; Torre-Cruz et al., 2022) approaches of heart health status detection have been reported. The unsupervised methods, despite their potential, have some limitations about their usability and accuracy (Safara et al., 2012). Since a labelled dataset is at hand (PhysioNet 2016), the proposed work focuses on a supervised method of depicting cardiac health status.

In the literature, several feature extraction methods, such as short-time modified Hilbert transform (STMHT) (Sun et al., 2014), short-time Fourier transform (STFT) (Soeta & Bito, 2015), Wigner-Ville distribution (WVD) (Obaidat, 1993), wavelet scattering transform (WST) (Mei et al., 2021), dual symmetric tree pattern (DSTP) (Barua et al., 2022), discrete wavelet transform (DWT) (Yuenyong et al., 2011), and continuous wavelet transform (CWT) (Debbal & Bereksi-Reguig, 2013) have been explored. The wavelet transform has been used more commonly in the literature as compared to other transform methods (Dwivedi et al., 2018) because it captures the signals effectively where the duration of the temporal window may be modified to facilitate multi-resolution analysis across a broad frequency spectrum (Zeng & Zhan, 2022). However, only a limited number of research works are available where textural-based features have been retrieved from 2-D images extracted from PCG signals.

1.1 | Major contribution

The lub-dub sound is produced as a heart valve shut and open. The characteristics of these lub-dub sounds vary for each cardiac functioning status, that is, normal, or abnormal. A comprehensive study of the characteristics of these signals is required to differentiate between a normal heart sound signal and an abnormal heart sound signal. Time-frequency domain analysis of these non-stationary heart sound signals has been conducted far and away using the spectrogram, mel-spectrogram and scalogram representation. The researchers have not yet been able to produce visuals from cardiac sound data that extract the pitch perception of a PCG signal. This work contributes in a significant way as follows:

1. This study proposes a unique approach for the identification of chroma and tone height characteristics in a PCG signal with respect to time using a time-chroma domain depiction known as chromagram which maps the spectral heart sound information into one octave. Each octave is segmented into 12 bins indicating each semitone.
2. A time-chroma representation of a PCG signal has the advantage of being insensitive to changes in timbre while still accurately portraying the signal's harmonic properties.

3. The arrangement of intensities and intriguing qualities of an image is described by the texture of image. Textural characteristics have already been employed in various applications like remote sensing, medical image processing, and image retrieval based on content-based images because of their significance in image processing. It has been observed through the review of literature that only a limited number of studies have been conducted to explore the extraction of textural features from the visual representation of PCG signals. However, this work further explores the sub-domain related to the heart sound signal classification in conjunction with the textural features as extracted from the chromagram.

1.2 | Machine learning in heart disease detection

During the past decade, heart disease was found to be most fatal all over the world. Cardiovascular diseases (CVDs) caused 17.5 million deaths in 2012, accounting for 31% of total deaths worldwide. Coronary heart disease caused an estimated 7.4 million deaths, and stroke in the second place caused 6.7 million deaths (World Health Organization, 2017). Majority of the deaths from CVDs occurred among low- and middle-income people. CVDs are preventable to a significant extent, if properly managed during the early stages of diagnosis. Thus, there is a dire need to improve the technologies capable of monitoring and analysing cardiac function parameters quickly and cost-effectively. The advancement of information technologies has aroused an expanding and ever-increasing interest in tracking the cardiovascular system for prolonged periods. This is being done as a potential approach to both diagnosing and managing cardiac conditions. Many new hidden patterns and similarities about the disease can be found through the use of machine learning algorithms. These algorithms used in heart sound classification, as shown in Figure 2, include neural network, decision tree (DT), SVM, Bayes net classifier, k-nearest neighbour (k-NN), hidden Markov model (HMM), and Gaussian mixture model (GMM) (Dwivedi et al., 2018).

As seen above, Section 1 introduced us to the various aspects and issues of the problem under study, and the contribution of proposed study in the heart sound classification domain. Section 2 demonstrates the theoretical foundations of the aforementioned domain. Section 3 reviews the relevant literature available on the problem under investigation. Section 4 illustrates the research methodology used in this work for an automatic cardiac activity status detection system. The results obtained through the experiments are presented in Section 5, while Section 6 draws the conclusion and describes the scope for further research.

2 | THEORETICAL BACKGROUND

2.1 | Human heartbeat analysis through PCG signal

Auscultation is the most commonly used and economical method for obtaining clinical information about heart valves (Hanna & Silverman, 2002). In order to make a diagnosis, physicians listen to the sounds that heart valves make with the stethoscope. While it is well established that humans

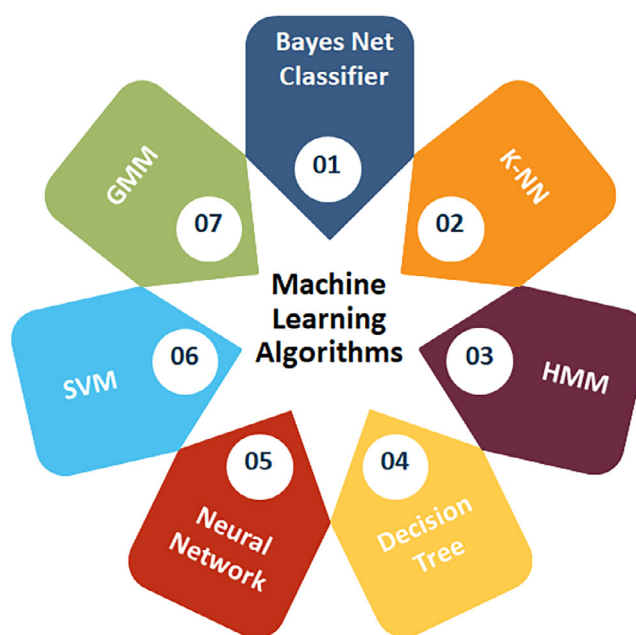


FIGURE 2 Machine learning algorithms used in heart sound classification (Dwivedi et al., 2018).

have a difficult time diagnosing heart murmurs through human hearing, the study of heart sounds and human hearing has confirmed that the diagnosis using the human ear is inadequate for cardiac auscultation (Jiang & Choi, 2006; Mangione & Nieman, 1997). Thus, heart sounds, which are made up of quantitative and qualitative aspects, are subject to human perception and skill. PCG, a graphic record of heart sounds, illustrates the mechanical energy generated by various cardiac components and processes (Durand & Pibarot, 1995). Due to this, an abnormal heart valve produces an abnormal PCG signal. A PCG signal consists of two heartbeat components which are referred to as 'lub-dub' sounds. The 'lub' sound generates S1, while 'dub' sound generates S2 as shown in Figure 3. S1 is generated when the closure of atrioventricular valves occurs at the start point of systole, whereas S2 is generated at the closure of pulmonary and aortic valves at the end of systole.

2.2 | Advantages of SVM over peer-models

SVM is a supervised learning model used to classify heart sounds using different kernel functions. Cortes and Vapnik were the first to report the implementation of an SVM on non-separable training data (Cortes & Vapnik, 1995). SVM classifiers were proven to be more efficient than artificial neural networks (ANNs) at classifying innocent murmurs (Kang et al., 2016). In terms of classification accuracy, neural networks performed admirably, but a large amount of training dataset is required to train neural networks. Further, classification is harder to accomplish with an ANN classifier because its mechanism seeks a large number of computational resources such as power and time (Dwivedi et al., 2018). Due to random selection of initial weights, back propagation ANNs (BP-ANNs) are unable to provide a viable approach for a classification task (Zheng et al., 2015). For high-dimensional classification problems, SVM is effective even when the dataset is small (Springer et al., 2015; Turkoglu et al., 2003). Further, the SVM classifier's performance is not directly linked to the number of dimensions of the input vectors (Ari et al., 2010). Whereas, HMM classifiers have a huge number of parameters; and the accuracy of classification is dependent on the specific HMM parameters that are used within the model (Fahad et al., 2018). Also, the SVM classifiers allow for the use of an optimum kernel function, which is either linear, Gaussian, polynomial, radial basis, Morlet wavelet kernel, Mexican hat wavelet, or b-spline, to be selected depending on the type of cardiac abnormality under investigation. The kernel function's parameters can be fine-tuned to enhance training reliability and produce the best possible performance of the model. Another drawback of HMM is that it takes a long time to process. k-NN additionally demands substantial memory resources (Wang et al., 2017), and it offers poor estimation and exceptional robustness to noisy training data (Moukadem et al., 2013). Deep learning has provided better results in the computer-aided classification of heart sound signals, but it also faces certain challenges. First of all, there are too many parameters of the deep learning model, with a large amount of data to be optimized, a long execution time, and a large training data set is required. Secondly, as the deep learning modelling calls for higher configuration of the computer with powerful CPU and GPU for calculation, the experiment cost runs high; and the model does not suit the home computers and microcomputers. However, the portable heart sound devices have great development potential and good application prospects (Li et al., 2020).

2.3 | Machine learning for annotating heart health status: Generic steps

Generally, an automatic heart sound classification system is decomposed into four stages: an audio signal is first captured and pre-processed, followed by segmentation of signals, then the features of the sound are extracted, and finally, a classification into normal and abnormal heart sound is done as shown in Figure 4 (Dwivedi et al., 2018).

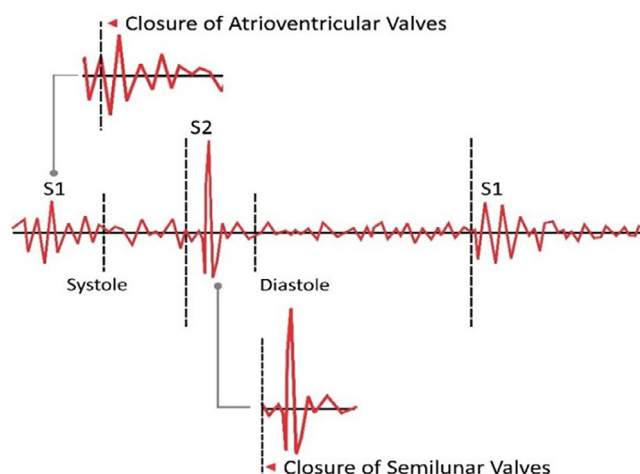


FIGURE 3 Representation of phonocardiogram (PCG) signal.

Segmentation-based PCG features have some drawbacks despite improvement in classification performance. In some cases, segmentation of PCG requires ECG recordings as well, which is hard to obtain. There are cases where abnormal peaks in signals are created due to murmurs, extra heart sounds are missed. This may also occur in patients where diastole is assumed to be longer than systole; patients exhibit non-informal murmurs (Singh et al., 2019). Unsegmented PCG recording classification provides a faster response with efficient results. There is a faster pace of system development when it is based on unsegmented cardiac sound classification (Zhang & Han, 2017). Deng and Han, in their work extracted features from PCG without executing the segmentation process and then proceeded to conduct a performance analysis using these features; and the findings were similar to those after segmentation of the heart sound (Deng & Han, 2016).

In several research works, the authors have extracted various time, frequency, and time-frequency features from the PCG signal dataset as shown in Figure 5. Cepstral features derived from audio files have been widely used in many sound detection applications. These are successful in structured environments, but have proven to be useless in the presence of noise (Kobayashi & Ye, 2014). However, features extracted from time-frequency images of speech or sound have proven useful in the presence of noise (Kobayashi & Ye, 2014). PCGs are also degraded by external noises (music, door close/open, etc.) and physiological noises (cough, speech, breathing sounds, lung sounds, stomach growls, etc.) noise which lead to misinterpretation of the signal, thus, inaccurate diagnosis. One of the advantages of converting audio data into an image domain for further feature extraction is time and frequency isolation. This is an important consideration because traditional methods mainly focus on the frequency variable while neglecting to allow for time in certain situations. Though, this two-dimensional analysis may preserve both spectral and temporal details. Therefore, many researchers in the heart sound classification domain have converted PCG sound files into spectrogram and scalogram images, and then further classified them into normal and abnormal classes. Because of these intriguing properties, our feature extraction approach has been built on the robustness of fusion of chromagram and spectrogram image descriptors. Chromagram is widely used in chord recognition systems, audio matching algorithms, and audio structure analysis (Yu et al., 2010). The chromagram is a more streamlined representation of the signal that places focus on the harmonic information. The use of chromagram allows for reduced computing costs and time as compared to other image descriptors (Bartsch & Wakefield, 2005).

3 | LITERATURE REVIEW

The literature review has been narrowed down to the domain of heart sound classification using an SVM classifier, by transforming audio data into image domain as shown in Figure 6. The three types of image descriptors applied in the reviewed literature are spectrograms, scalograms, and mel-spectrograms.

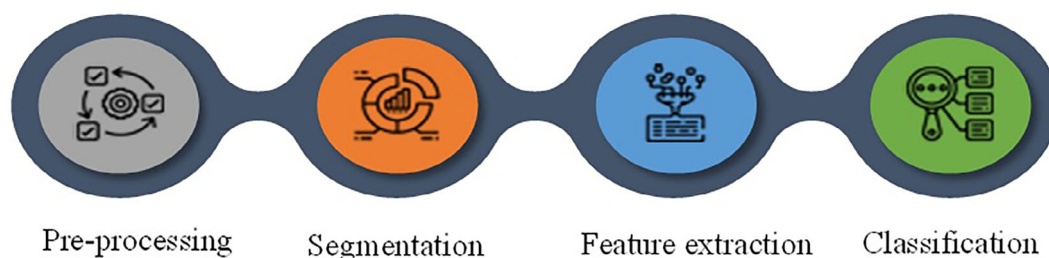


FIGURE 4 Stages of automatic heart disease classification system (Dwivedi et al., 2018).

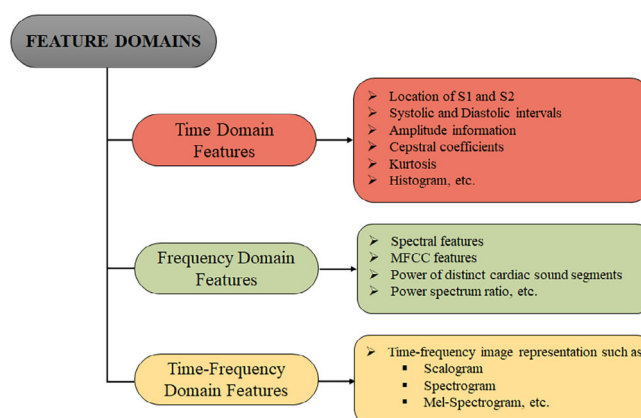


FIGURE 5 Types of features extracted from phonocardiogram (PCG) signals.

3.1 | Spectrogram

A spectrogram is a thorough representation of audio that can display the time, frequency, and amplitude of a signal on a single graph. It shows how the frequencies of a signal change over time. The third dimension is then used to represent amplitude, which can be depicted with varying brightness or colours as mentioned in Figure 6a. Fourier transforms are used to create spectrograms from audio signals (Zhang et al., 2017a, 2017b). The amplitude of each frequency included in the signal is displayed using a Fourier transform, which decomposes the signal into its constituent frequencies. A spectrogram divides the duration of audio into smaller time segments and uses the Fourier transforms to detect the frequencies contained within each segment. The Fourier transforms for all of those segments are then combined into a single graphic. It shows frequency (Y-axis) versus time (X-axis), with different colours indicating the frequency's amplitude. Higher the signal's energy, brighter would be the hue.

In 2020, Khalid et al. made a comparative analysis of various classifiers namely, SVM, DT, ANN, random forest, naive Bayes, and k-NN in their proposed model by extracting mel-frequency cepstrum coefficients (MFCC), delta MFCC and FBANK features with the respective accuracy of 56.00%, 27.60% and 33.10% after using the SVM classifier (Nahar et al., 2020). In this proposed work, the researchers have used a small amount of dataset for training the model which led to overfitting or underfitting issue.

In 2019, researchers (Ferdoushi et al., 2019) used PASCAL Classifying Heart Sounds Challenge 2011 database, clipped all PCG signals to 5 s from the data set, and computed the STFT after sampling the signal at the rate of 22,050 Hz. The time-domain features such as zero crossings were extracted, while the rest were frequency domain features such as spectral roll-off, spectral bandwidth, MFCC, and spectral centroids obtained from STFT. Time series was acquired by calculating the average of the extracted spectral features for 512 samples. Training data was used to train the linear SVM classifier using these features. The authors used test data set for detecting various diseases by applying the proposed model and calculated the accuracy score for performance evaluation of the proposed method. In their study, (Dong et al., 2019) mentioned two major features of fusion strategies: early fusion and late fusion. The machine learning model in the early fusion scenario was fed features that were concatenated beforehand. Late fusion, on the other hand, was applied by training the models separately with different feature sets and

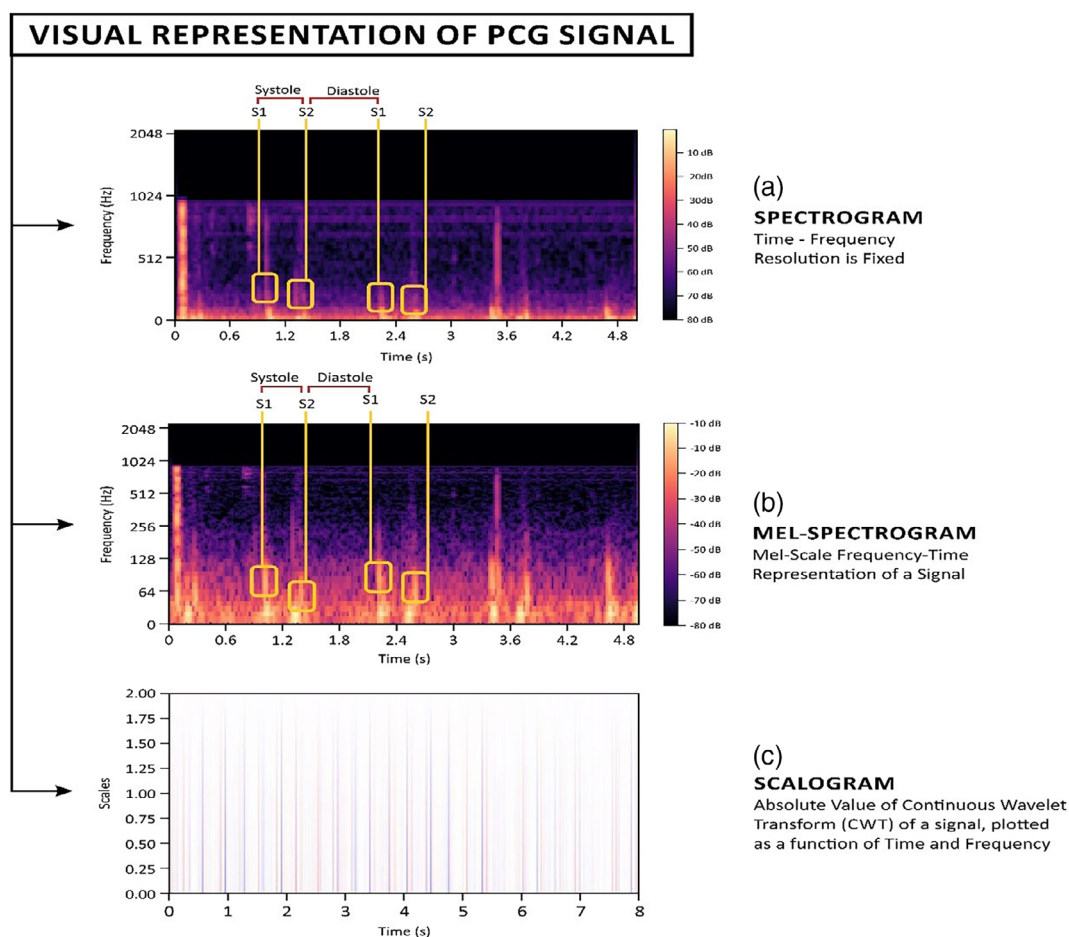


FIGURE 6 Image descriptors applied in heart sound representation.

making the final prediction using a voting process. The authors implemented two late fusion techniques in their research, namely, majority voting (MV), and margin sampling voting (MSV). The final prediction for MV was given to the one that received maximum votes from the individually qualified classifiers. The final MSV prediction was determined by the classifier with the highest margin sampling value, which was calculated as the difference between the first and second highest probability values. The analysis showed that in comparison to MFCC and formant-based features (F1–F3), the performance of log mel feature was outstanding. The heart sound structure, in this study, was benchmarked by the SVM model trained by log mel with a uniform average recall (UAR) of 49.70%.

In 2017, (Zhang et al., 2017b) proposed a heart sound classification system in which the spectrograms of the segmented heart cycles were first scaled to a fixed size. The scaled spectrograms were then subjected to a dimension reduction procedure called principal component analysis (PCA) to remove the most discriminative elements. The underlying structure of the scaled spectrograms was derived using the tensor decomposition method during the dimension reduction step. Finally, the SVM algorithm was applied to complete the classification task. The suggested procedure was also tested on three publicly available databases from the PASCAL Classifying Heart Sounds Challenge 2011 and the PhysioNet 2016. The result metrics used to represent the classification model's result were precision for normal heart sounds (P_n) and Precision for abnormal heart sounds (P_{ab}) as shown in Table 1. In their work (Zhang et al., 2017a), proposed another heart sound classification system using scaled spectrograms for feature extraction and applied partial least squares regression (PLSR) for dimension reduction to eliminate redundant features. Finally, SVM was applied for the classification process on public datasets provided by PASCAL Classifying Heart Sounds Challenge 2011. The result metrics used to represent the classification model's result were precision for normal heart sounds (P_n), precision for murmurs (P_m), precision for artefacts ($P_{artifact}$), precision for extra heart sounds (P_{extra}), specificity, and sensitivity as shown in Table 1. In their research work Faturrahman et al. applied the Shannon energy envelope (SEE) technique for segmentation of PCG signals, and further extracted spectrogram by applying the STFT algorithm (Faturrahman et al., 2017). Since the spectrograms contain redundant data, the researchers applied PCA for dimension reduction and then used deep belief network (DBN) for feature extraction, followed by the application of SVM algorithm for classification. All the heart sound classification systems as mentioned above included PCG signal segmentation as a part of their process cycle. Heart sound segmentation is a difficult problem due to the non-stationary nature of the heart sound signal and the existence of background noise in the real-world including murmurs and other artificial noises in the real-world. In fact, without heart sound segmentation, the basic task of heart sound classification would be possible. The primary task's goal is to simply detect whether a heart disorder is present in the signal rather than to further identify it which necessitates pinpointing the exact location of the heart disease in the signal. Zhang et al. have also mentioned the same in their work (Zhang & Han, 2017) and proposed a heart sound classification system without segmentation by extracting spectrograms from PCG sound data, scaling them to a fixed size using bilinear interpolation method, and then using convolutional neural network (CNN) and SVM algorithms for feature extraction and classification, respectively. The performance metric for the classification model proposed in this work is unweighted average recall (UAR) as shown in Table 1.

Nilanon et al. proposed a heart sound classification system in which classifiers were trained to classify the segments after windowing the recordings into 5-s segments (Nilanon et al., 2016). The overall classification of the recording was then developed using a voting scheme based on the classification results of its parts. Spectrograms and MFCC were the features extracted in this work.

3.2 | Mel-spectrogram

A mel-spectrogram differs from a conventional spectrogram as it plots frequency versus time in the following two ways:

1. On the y-axis, the mel-scale is used instead of frequency.
2. Colours are indicated using the decibel scale rather than the amplitude scale.

Recently, Yildirim extracted spectrogram images using mel-spectrogram method from PCG signals and then inserted the feature maps obtained through the Darknet53 architecture technique as an input in various classifiers for the comparison purpose (Yildirim, 2022). One of the data augmentation methods, that is, interpolation method was also used for training the proposed classification model. This hybrid classification model achieved 99.63% and 97.19% accuracy for two different subsets of the dataset used in this work. The work produced commendable results but it lacked to provide any information about the structural schema of the deep learning model employed in this context.

Chowdhury et al. also proposed a hybrid model of deep learning and machine learning algorithms for PCG signal classification domain. The PCG signals were pre-processed using hamming window technique (Chowdhury et al., 2021). The authors created power spectrum using mel-spectrogram and MFCC. The output of CNN with rectified linear unit (ReLU) activation function served as input features to the machine learning algorithms used as classifiers to predict the PCG signal status. A comparative analysis of machine learning model and hybrid deep learning model was also presented in this work, where, only SVM appeared with the accuracy of 79.56%, while CNN-SVM resulted into 92.10% accuracy. The disadvantage of using the deep learning algorithm with machine learning algorithm as a hybrid model was that it increased its training time and made the computing more complex.

TABLE 1 Results of peer research works.

Citation	Dataset	Image type	Features	Results (in %)					
				P	S _e	S _p	Acc	UAR	
(Zhang et al., 2017b)	X _A	Spectrogram	Tensor features	n = 100, ab = 64	-	-	-	-	-
	X _B			n = 51, ab = 95	-	-	-	-	-
	Y			n = 92, ab = 88	-	-	-	-	-
(Nahar et al., 2020)	U		MFCC, Delta MFCC and FBANK	57.2	55.1	-	55.1	-	-
(Zhan et al., 2017a)	X _A		Regression features	n = 67, ab = 91	100	64	-	-	-
	X _B			n = 77, ab = 66	51	90	-	-	-
(Nilanon et al., 2016)	Y		Power spectral domain features and MFCC features	-	75.6	75.6	-	-	-
(Ferdoushi et al., 2019)	X _A	Spectrogram	Spectral centroid, spectral roll-off, spectral bandwidth, zero crossings, and 20 MFCC coefficients	-	-	-	72.2	-	-
	X _B			-	-	-	72.2	-	-
(Zhang & Han, 2017)	Z		ComParE acoustic feature Set	-	-	-	-	47.9	-
(Dong et al., 2019)			MFCC, log-mel, eGe MAPS, F3-F3	-	-	-	-	49.7	-
(Faturrahman et al., 2017)	W		Deep belief network (DBN) feature extraction method	-	-	-	86	-	-
(Yildirim, 2022)	U	Mel-spectrogram	Feature maps obtained by applying the Darknet53 architecture	98	98	99	97	-	-
(Chowdhury et al., 2021)	Y		Feature maps obtained as output of CNN	-	93.3	91.8	92.1	-	-
(Humayun et al., 2018)	Z + Y		ComParE features + auto-encoder features	-	-	-	64.2	42	-
(Amiriparian et al., 2018)	Z	Mel-spectrogram	ComParE feature set and BoAW features	-	-	-	-	47.9	-
(Meintjes et al., 2018)	Y	Scalogram	CNN features	-	87.7	86	85.9	-	-
(Alaskar et al., 2019)			Basic image features such as blobs, edges, and so forth extracted by AlexNet	-	83.7	90	87.6	-	-
Proposed work	Y	Chromagram	Textural features: LBP, ALBP, RLBP	93.11	97.53	68.59	92.12	-	-

Abbreviations: ab, precision for abnormal class; Acc, accuracy; n, precision for normal class; P, precision; S_e, sensitivity; S_p, specificity; U, Kaggle heartbeat sounds; UAR, unweighted average recall; W, Massachusetts Institute of Technology Heart Sound Database (MITHSDB) and Aalborg University Heart Sound Database (AADHSDB); X_A, PASCAL Classifying Heart Sounds Challenge 2011 dataset A; X_B, PASCAL Classifying Heart Sounds Challenge 2011 dataset B; Y, PhysioNet 2016; Z, INTERSPEECH 2018 ComParE HSS dataset.



Humayun et al. in 2018, proposed an ensemble of classifiers for the INTERSPEECH 2016 Computational Paralinguistics Challenge feature set (ComParE) to automatically detect irregular cardiac sounds of various intensity ranges (Humayun et al., 2018). The main architecture was built around parameter transfer learning from a 1-D CNN model that had been pre-trained on the PhysioNet 2016 dataset. The authors also used a deep auto-encoder-based architecture to learn unsupervised function representations from mel-spectrograms; and finally, ComParE feature set was used comprising a range of prosodic, spectral, cepstral, and voice quality low-level descriptor features and an SVM classifier to build a segment-level feature-based framework. In their work, Amiriparian et al. firstly, derived mel-spectrograms from raw heartbeat recordings (Amiriparian et al., 2018). After that, a sequential auto-encoder was developed, trained on these extracted spectra, which were considered to be time-dependent frequency vector sequences. The learned mel-spectrogram representation of PCG signals was then constructed to be used as feature vectors for the corresponding contexts after auto-encoder training. Finally, the ComParE and bag-of-audio-words (BoAW) feature sets were used to train an SVM classifier which predicted the labels of the heartbeat recordings.

3.3 | Scalogram

The scalogram is a depiction showing the absolute value of a signal's CWT as a function of time and frequency. It should be used when there is a requirement of better time localization for a short duration, high-frequency events, and better frequency localization for long-duration, low-frequency events.

In 2019, Alaskar et al. firstly, pre-processed every subject's raw PCG signals (Alaskar et al., 2019). Filtering signals were used to minimize noise because the background noises could distort and harm the recording signals. The heart sound components were then detected through the segmentation process. Further, scalogram images of PCG signals were produced by using CWT after segmenting the signals. Finally, two methods were tested: in the first method, pre-trained AlexNet based on CNN was used to extract features, which were then fed on an SVM classifier. Whereas, in the second method AlexNet was used for feature extraction as well as a learning classifier. In all, 9216 features were extracted in the pre-trained AlexNet method which was further fed as training input to the SVM classifier. The SVM classification of features derived from the pre-trained AlexNet outperformed the AlexNet end-to-end learning classifier.

In 2018, Meintjes et al. examined the classification of fundamental heart sounds using CWT scalograms and CNN. The classifiers like CNN, SVM, and k-NN were contrasted in their classification of the first and second heart sounds of scalograms generated by the Morse analytic wavelet (Meintjes et al., 2018). A freely accessible dataset of normal and abnormal heart sound data was used to collect samples of the first and second heart sounds; and magnitude scalograms were determined for each study. CNNs were trained and tested using these scalograms. The classification of features derived from the network's completely linked layer was compared to linear binary pattern (LBP) features. When it came to identifying between the first and second heart sounds, the CNN had an overall classification accuracy of 0.86. Related findings were obtained using features derived from the CNN and categorized using an SVM (0.859). The authors concluded that by using both SVM and k-NN classifiers, CNN features outperformed LBP features in classification.

After reviewing the literature on the use of an image corresponding to a PCG signal, it has been discovered that there is no information concerning pitch variations over time. The vital information regarding clicks and snap sounds produced throughout a cardiac cycle owing to the opening and closing of heart valves may be stored in pitch classes. It has also been observed through the literature review that during the period 1999–2022, there is hardly any mention of the use of chromagram representation of PCG signals for categorizing them into normal and abnormal.

4 | METHODOLOGY

Firstly, the visual representation through the fusion of chromagram and spectrogram image extraction was done to reflect the input audio signal. Then the textural features of both chromagram and spectrogram images were extracted using LBP and its variants called ALBP and RLBP. Finally, the SVM classifier was used to classify the heart sounds. Figure 7 provides a detailed outline of all the preceding measures. The PhysioNet 2016 heart sound database was used to report the experimental findings.

4.1 | Data acquisition

The dataset was collected from PhysioNet 2016 which contained heart sound recordings obtained from a variety of sources around the world, in clinical as well as non-clinical settings from both healthy and pathological patients. The challenge training collection included five databases (A–E) with a total of 3126 heart sound recordings ranging in length from 5 to 120 s (Goldberger et al., 2000). The heart sound recordings were taken from various auscultation locations of the body. The aortic region, pulmonic area, tricuspid area, and mitral area were the most common four auscultation sites to examine heart sounds. Two training classes, namely, normal and abnormal heart sound recordings, were created for

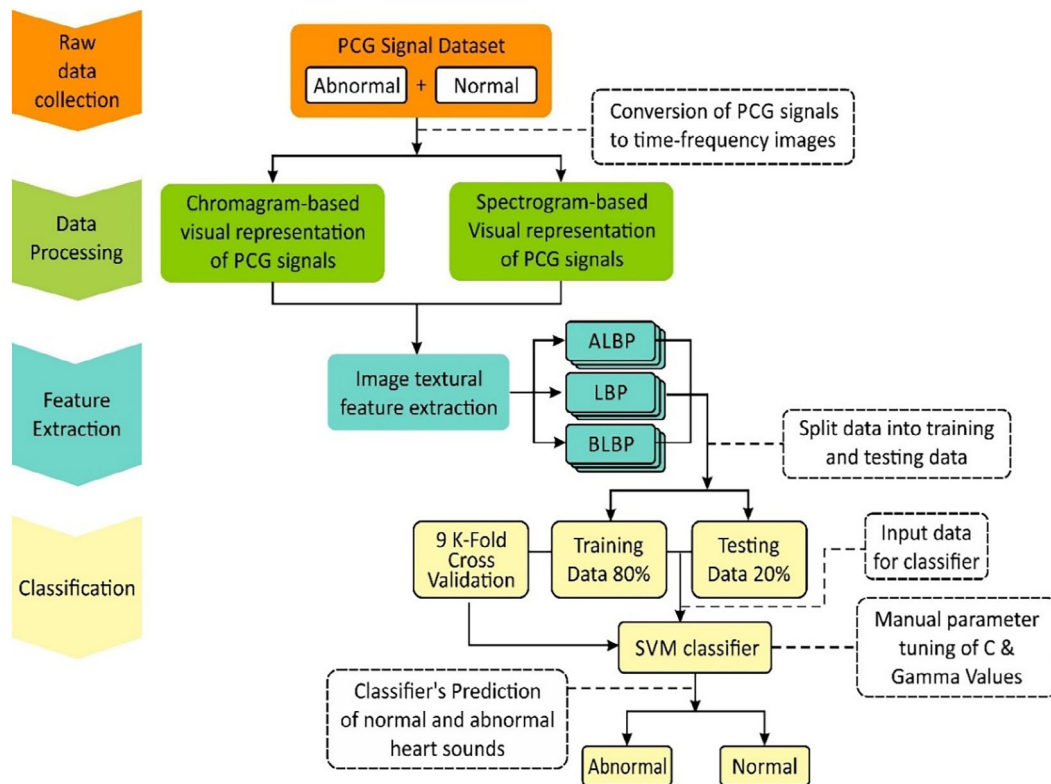


FIGURE 7 Proposed methodology of phonocardiogram (PCG) signal classification model.

both the training and test sets. The normal recordings pertained to healthy people, while the abnormal ones were recorded from those who reported heart conditions. Children and adults were included as both stable subjects and pathological patients. The heart sound recordings of each patient were contributed between one and six recordings. There were recordings of different lengths ranging from a few seconds to more than 100 s. All the recordings were resampled to 2000 Hz and were available in '.wav' format. There was only one PCG lead in each recording (Liu et al., 2016).

4.2 | Image-based visual representation of PCG signals

At this stage, the '.wav' format audio files were converted into chromagram and spectrogram images. Features were extracted using the chromagram and spectrogram-based visual representation of PCG signals. The extraction of audio features in the image domain was advantageous in many ways, including time and frequency independence, as well as more flexible temporal comparison. These considerations were quite significant because traditional methods placed a lot of focus on the frequency variable while neglecting to account for time in many cases. Instead, these methods produced a 2D description that could retain both spectral and temporal details (Birajdar & Patil, 2020). Although the time-frequency representation of a PCG signal is useful for depicting the signal, but it fails to incorporate the information about chroma and tone height features of a heart sound signal. Thus, this work introduces an alternative approach of heart sound signal representation that allows for a more precise measure of pitch-related changes in the heart sound.

The chroma vector is a 12-element representation of an audio signal's short-time energy distribution. A chromagram is the integration of these 12-dimensional chroma vectors over time, that is, spectral energy is mapped into spectral bins corresponding to the 12 elements of the chroma vector. Chromagram is represented as the time-chroma distribution of a signal; it is a representation of joint signal strength distribution over the variable's time and chroma (Bartsch & Wakefield, 2005). Shepard et al. proposed pitch perception based on tone height and chroma in which the researchers modelled human pitch vision as a helix with a 1-D line (Shepard, 1964). The vertical dimension expresses the continuous tone height, while the angular dimension expresses the chroma. According to Bartsch and Wakefield (2005), pitch (p), which was a value in Hz, was expressed using chroma (c) and tone height (h):

$$f_p = 2^{c+h} \quad (1)$$

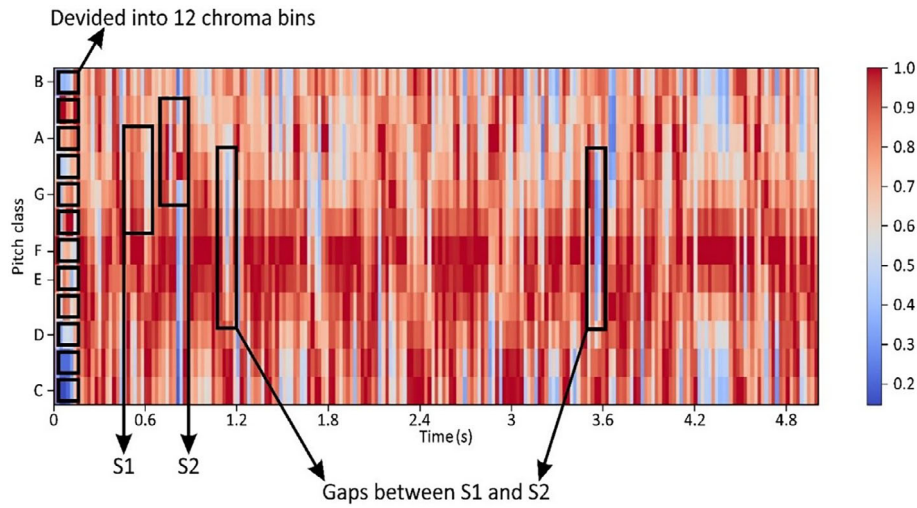


FIGURE 8 Chromagram image of a heart sound signal.

where, f stands for frequency; and c belongs to set $[0,1]$. The chroma vector uses the octave invariance principle which states that audio signals divided by doubling of frequency have no functional difference. The chroma vector is calculated by dividing a short-term window's STFT coefficients into 12 bins. A single bin is represented as a single pitch class out of 12 equal-tempered pitch groups. The chroma bin's i th chroma vector as given by Sell and Clark (2014) is as follows:

$$C(i) = \sum_{r=0}^{R-1} \left| X \left(2^{i+\frac{r}{12}} f_{\min} \right) \right| \quad (2)$$

where, l represent the total number of bins in the chroma vector; and $f_{\min}$ is the lowest frequency. The chroma computation is performed over the R number of octaves. The chromagram displays energy for each chroma vector against time. For different energy values, a chromagram visual representation in the form of a 2-D image is a useful tool in audio signal classification. During the chromagram computation, an FFT window size of 1024 with a 512-hop length was used in this work. Figure 8 presents the 12 bin chromagram image of one of the PCG signals derived from the PhysioNet 2016 database.

4.3 | Extraction of visual textural features

The original PCG signal was interpreted in the visual domain to perform visual feature extraction. For this purpose, a 2-D image of the chromagram and spectrogram was generated from the input PCG signal. LBP features were extracted from these chromagram and spectrogram images because these were the simplest and most efficient textural descriptors (Ren et al., 2016; Yang & Krishnan, 2017). The LBP descriptors effectively captured variations in chroma dynamics in the chromagram representation of the audio signal. The chroma bin transformations, in addition to the edges, were recorded and encoded using the relative intensities of neighbour pixels, which were resistant to monotonic grayscale variations (VenkateswarLal et al., 2019).

LBP is used in image processing applications due to its resistance to grayscale variance and low computational complexity. The textures represented by the 2-D chromagram image are found to be relevant for the extraction of LBP features in the speech/music classification domain (Birajdar & Patil, 2020). Due to this reason, it has been applied in the domain of heart sound classification. The LBP model defines the texture by taking into account each pixel c , P neighbours evenly spaced at a distance of R as shown in Figure 9. Using the position of a centre pixel (x,y) as a cut-off value and the radius of the neighbourhood as R , the LBP operator has been obtained by finding a correlation between c and p neighbours; where, $p = 1, 2, \dots, P$ and P is the number of neighbours according to Equations (3) and (4) (Ojala et al., 2002).

$$\text{LBP}(x,y) = \sum_{p=0}^{P-1} s(g_p - g_c) 2^p, \quad (3)$$

$$s(g_p - g_c) = \begin{cases} 1, & g_p - g_c \geq 0 \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

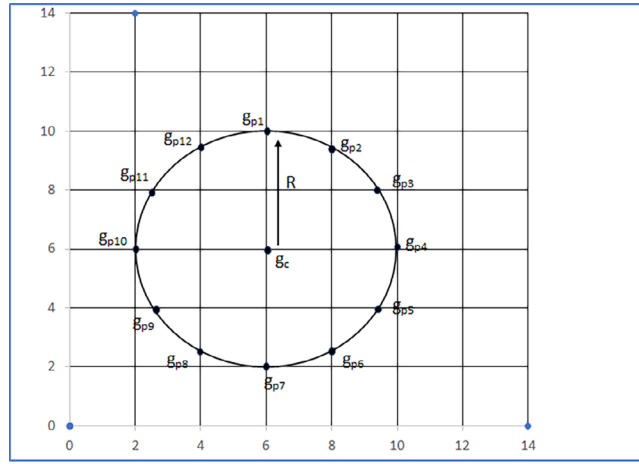


FIGURE 9 Representation of linear binary pattern (LBP) feature points on an image.

where, $g(c)$ and $g(p)$ denote central pixel intensity and neighbouring pixel intensity, respectively.

Based on the transformation numbers between 0 and 1 s in the LBP picture (Birajdar & Patil, 2020), a uniform LBP concept was proposed. If the number of 0–1 transitions is less than or equal to 2, the LBP is considered a uniform pattern. Since 10010101 include six transformations, the code is considered non-uniform and 01000000 is considered uniform as it contains two transitions. LBP with non-uniform patterns is stored in an extra bin of the histogram, while the instances of the potential uniform pattern are stored in a uniform histogram bin. $LBP_{p,R}^u$, where R is the radius; and P is the number of neighbourhood pixels around the central pixel (Equations (5) and (6)).

$$LBP_{p,R}^u(x,y) = \begin{cases} \sum_{p=1}^{P-1} s, & \text{if } U(LBP_{p,R}^u(x,y)) \leq 2 \\ P+1, & \text{otherwise} \end{cases} \quad (5)$$

$$U(LBP_{p,R}^u(x,y)) = |s(g_{p-1} - g_c) - s(g_0 - g_c)| + \sum_{p=1}^{P-1} |s(g_p - g_c) - s(g_{p-1} - g_c)| \quad (6)$$

$[(P-1)P+2]$ binary patterns, in all, are constructed in uniform LBP patterns. Despite the fact that the number of patterns generated by uniform LBP is $[(P-1)P+2]$, which is less than the LBP that produces 2^P patterns, and it still contains useful information for classification. Uniform LBP provides a function descriptor with a much smaller number of features, that is, 59 features without using the entire histogram with 256 features. This 59-dimensional feature vector can be used to represent the chromagram and spectrogram image. During the tests, it was discovered that uniform LBP feature extraction produced high classification accuracy. As a result, image zoning was not necessary to leverage local features, saving computational time and resulting in a small feature vector dimension.

To enhance the performance of classification, the directional statistical feature vector can be used by minimizing the mean and SDs of directional differences along different orientations. Keeping it in view, Guo et al. proposed ALBP which contributed towards minimizing the estimation error that occurred as a result of calculating the local difference between each pixel and its neighbours (Guo et al., 2010). A new parameter called weight (w_p) is defined in the LBP equation; and the definition of ALBP is demonstrated in Equation (7).

$$ALBP_{p,R} = \sum_{p=0}^{P-1} 2^p s(g_p \times w_p - g_c) \quad (7)$$

Here, w refers to the weight vector of ALBP; and each w_p is calculated along with one orientation, that is, $2\pi p/P$ for the whole image.

In RLBP, the intensity of the central pixel is compared with the average intensities of neighbouring pixels present in a ring-shaped area around the centre pixel. Mathematical representation of RLBP is explained in Equations (8) and (9) as follows:

$$RLBP_{p,R,N} = \sum_{i=0}^N \text{sign}(g_i - g_c) 2^i, \quad (8)$$

$$g_i = \left(\sum_{p=0}^{P-1} \sum_{r=i \times R}^{(i+1) \times R} g_{p,r} \right) / RP \quad (9)$$

where, the index of i th ring between radius $(i \times R)$ and $[(i+1) \times R]$ around the centre pixel is represented by i . The difference between the radius of two adjacent rings is represented by R ; and the number of rings around the centre pixel are represented by N .

4.4 | Classification using SVM classifier

SVM is a non-probabilistic, powerful supervised learning model for classification and regression analysis that is focused on structural risk minimization (Cortes & Vapnik, 1995). It is widely used in pattern classification problems (Akram et al., 2018; Prabukumar et al., 2019), and classifying the input signal into two groups; normal and abnormal as used in this paper.

Considering D_T as the training data with n points dataset and x_i as the input data labelled as $y_i = 1, -1$ for two classes respectively.

$$D_T = \{(x_i, y_i) | x_i \in R^n, y_i \in (\pm 1)\}_{i=1}^n \quad (10)$$

The n -dimensional hyperplane created by the SVM classifier can be depicted as in Equation (10) and also through Figure 10.

$$f(x) = w \cdot x + b = \sum_{i=1}^n w_i x_i + b \quad (11)$$

Here, x is a $n \times 1$ vector while w and b are normal and scalar values respectively to the hyperplane. Hyperplane can be expressed as:

$$w \cdot x_i + b = \pm 1 \quad (12)$$

An ideal hyperplane creates the maximum separation boundary, and the SVM searches for the optimal hyperplane that satisfies the performance criteria of the model. For x_i input pattern in the training data, the linear SVM classifier operation is described by w, b that complies with the following stipulations:

$$\begin{cases} w_i x_i + b > +1, & \text{if } y_i = +1 \\ w_i x_i + b < -1, & \text{if } y_i = -1 \end{cases} \quad (13)$$

The input data x in SVM can be classified as:

$$f(x) = \sum_{i=1}^n a_i y_i k(x, x_i) + b \quad (14)$$

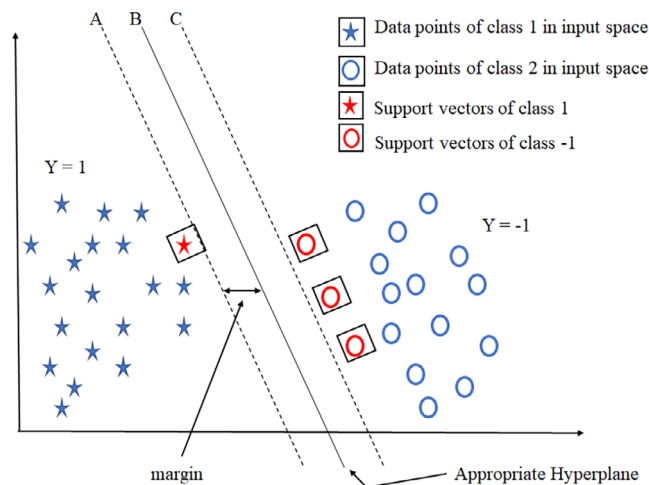


FIGURE 10 Representation of support vector machine (SVM) hyperplane separating binary classes on an input space.

Here, class labels are represented by y_i , that is, $\{1, -1\}$; x_i is the input training data; k is the kernel function; and a_i and b are the model parameters. In this work, grid search technique has been used to set the values of C , gamma and kernel. Range values given for each parameter are: $C = [1-10]$, gamma = $[0.01-1]$, and kernel = [polynomial, RBF, linear, cubic]. The most reliable results were obtained with the values of $C = 3$, gamma = 0.01, and kernel = RBF.

The features generated after implementing the LBP feature extraction technique over chromagram images of PCG signals are fed to SVM classifier for heart sound classification.

The input points were mapped to a high-dimensional feature space; and a maximum margined hyperplane was created to separate the points pertaining to two different classes when SVM algorithm was applied for the binary classification model. However, in several applications, some input points could not be precisely assigned to either of these two classes. Some of these points should be assigned entirely to one class for enabling the SVM to differentiate these points more accurately. Some noise-corrupted data points are less useful; thus, the system should discard them. In this research work, a fuzzy membership has been applied to each input point of SVM so that various input points can contribute differently to the learning of class separating hyperplane. This method was initially proposed by Lin et al. in their research work titled as 'fuzzy support vector machines' (Lin & Wang, 2002), where the authors pointed out that FSVM improved the SVM by lowering the impact of data point outliers and disturbances. FSVM is quite appropriate for applications involving data points with unmodelled properties. However the present research work has also implemented the least square SVM (LSSVM) proposed by Suykens et al. where a set of linear equations has been used to train the classification model instead of quadratic optimization problem which is availed in the traditional SVM algorithm (Suykens & Vandewalle, 1999). Khemchandani et al. introduced the twin support vector machine (TWSVM), a novel non-parallel plane classifier for binary classification models. TWSVMs also sought to generate two non-parallel planes so that each plane could be as close as possible to one of the two classes and as far as possible from the other (Khemchandani & Chandra, 2007). The concept of TWSVMs was quite similar to that of traditional SVMs. However, TWSVMs were fundamentally different from SVMs. The TWSVM algorithm solved two quadratic programming problems (QPPs), whereas the SVM algorithm solved only one QPP. In traditional SVMs, the QPP contains all input data points in the constraints, but in TWSVMs, the constraints are dispersed so that each class pattern provides the constraints of the other QPP. This method of solving two small-sized QPPs as against the one large-sized QPP enabled TWSVMs to function more speedily than the traditional SVMs.

For the purpose of current research work, all the three advance-SVM algorithms, as mentioned above, were applied as a classifier in the proposed PCG signal classification model for having a comparative analysis with traditional SVM classifier. The results have been discussed in Section 5 of this paper, which confirm that performance of the traditional SVM algorithm is far better than other algorithms.

5 | EXPERIMENTS AND RESULTS

5.1 | Hardware and software specifications

All the experiments were run on a computer system with an Intel core i9 processor running at 4.30 GHz using 32 GB of RAM and having 64-bit operating system. The algorithms of signal pre-processing (PCG signal to image conversion), feature extraction, and classification were all implemented using Python (version 3.9) on scikit-learn (version 1.1.1) and scikit-image (version 0.18.1) libraries among others.

In the current work, chromagram and spectrogram images from heart sound signals were extracted through the LBP feature extraction technique. ALBP and RLBP features were also extracted, and a comparison was undertaken for analysing the significance of these features with LBP. The LBP features provided better results as compared to the other two feature extraction methods. Further, the SVM algorithm was used for classification on the dataset with two classes (abnormal and normal). Boosting techniques (Ada-boost, Rus-boost, and XG-boost) in fusion with SVM for this purpose were also tried, but SVM without boosting performed in a better way as compared to boosting techniques. Some advance SVM algorithms, that is, fuzzy SVM, LSSVM, and twin SVM were also implemented in the proposed work for a comparative analysis. For the purpose of comparison among the various classification models that had been implemented, the performance evaluation of each model was done by calculating its accuracy, specificity, sensitivity, recall, precision, and F1-score as shown in Table 2 using Equations (15)–(19). Firstly, a confusion matrix was calculated to get all the above-mentioned performance evaluation scores of the classification model. A confusion matrix is a table that describes how well a classification model has performed on a set of test data for which the true values are to be known. The terminologies used in a confusion matrix are true positive (TP), true negative (TN), false positive (FP), and false negative (FN). Accuracy describes the ratio of correctly classified occurrences to all instances.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (15)$$

$$\text{Sensitivity} = \frac{TP}{TP + FN} \quad (16)$$

TABLE 2 Experimental results of SVM, SVM with boosting techniques, LSSVM, fuzzy SVM, twin SVM, decision tree, k-NN, and ANN deployed on LBP, ALBP, and RLBP feature set.

Features	Classification technique	Testing accuracy	Specificity	Sensitivity	Precision	F1-score
LBP	SVM (RBF kernel)	92.12	68.59	97.53	93.11	95.27
	SVM + ADA-boost	79.48	0	100	81.32	89.70
	SVM + RUS-boost	83.43	85.95	81.97	96.21	88.52
	SVM + XG-boost	79.44	0	100	81.32	89.70
	LSSVM (RBF kernel)	89.04	65.18	95.32	91.23	93.23
	Fuzzy SVM (polynomial kernel)	79.78	0	100	79.78	88.75
	Twin SVM	80.09	0	100	80.09	88.94
	Decision tree	84.46	45.00	93.43	88.2	90.74
	k-NN	88.27	56.00	95.98	90.12	92.96
	ANN	90.74	63.97	97.07	91.92	94.43
ALBP	SVM (RBF kernel)	91.35	72.65	95.96	93.44	94.68
	SVM + ADA-boost	80.24	0	100	80.24	89.04
	SVM + RUS-boost	81.84	82.81	80.38	95.00	87.08
	SVM + XG-boost	79.88	0	100	80.24	89.04
	LSSVM (polynomial kernel)	88.73	72.59	92.98	92.80	92.89
	Fuzzy SVM (polynomial kernel)	79.32	0	100	79.32	88.46
	Twin SVM	79.00	0	100	62.00	69.00
	Decision tree	82.51	34.55	94.23	85.48	89.64
	k-NN	84.56	45.58	94.92	86.78	90.67
	ANN	91.35	67.72	97.06	92.57	94.76
RLBP	SVM (RBF kernel)	90.94	78.32	96.43	94.01	95.21
	SVM + ADA-boost	79.25	0	100	77.93	87.59
	SVM + RUS-boost	80.84	90.9	80.79	96.91	88.12
	SVM + XG-boost	80.09	0	100	77.93	87.59
	LSSVM (polynomial kernel)	88.11	61.48	95.12	90.37	92.68
	Fuzzy SVM (polynomial kernel)	79.01	0	100	79.01	88.27
	Twin SVM	80.24	0	100	80.24	89.04
	Decision tree	83.33	24.21	97.69	84.14	90.41
	k-NN	83.64	35.03	96.67	84.73	90.31
	ANN	90.53	70.53	95.94	92.32	94.10

Abbreviations: ALBP, adaptive local binary pattern; ANN, artificial neural network; k-NN, k-nearest neighbour; LBP, linear binary pattern; LSSVM, least square SVM; RLBP, ring local binary pattern; SVM, support vector machine.

The sensitivity measure denoted the proportion of patients with the disease who were also tested positive through the proposed model. Applying the same understanding, the sensitivity will be the model metric used to select the best model when there is a high cost associated with false negative. Whereas specificity denotes the proportion of patients having no disease; and the proposed model also tested them negatively.

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \quad (17)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (18)$$

Precision is the number of positive predictions divided by the total number of predicted positive class values. It is also known as the positive predictive value. In other words, it measures the exactness of a classifier. A large number of false positives indicate low precision. F1-score is a performance evaluation metric that takes both precision and recall (also known as sensitivity) values into account for better judgement of a classifier.

$$F1\ score = \frac{TP}{TP + \frac{1}{2(FP+FN)}} \quad (19)$$

The figures presented below (Figure 11a–c) display the histograms of all the results exhibited in Table 2. Figure 11a–c present a comparative analysis based on the metric measure performance of various classifiers after extracting the LBP, ALBP and RLBP features respectively. The accuracy as displayed in Table 2 and Figure 11a–c was achieved on the test dataset which was 20% separated from the entire dataset before training the models.

Figure 11a–c reflect the comparative performance of all the three feature extraction techniques used for the purpose of this research work. SVM with RBF kernel classifier and LBP feature extraction method has outperformed all other techniques. Out of the 30 models, some of the models, namely, LBP with SVM-AdaBoost fusion, LBP with SVM-XGBoost fusion, LBP with fuzzy SVM, LBP with twin SVM, ALBP with SVM-AdaBoost fusion, ALBP with SVM-XGBoost fusion, ALBP with fuzzy SVM, ALBP with twin SVM, RLBP with AdaBoost fusion, and RLBP with XGBoost fusion, have recorded 100% sensitivity, while the specificity has been 0% as reflected in Figure 11a–c and Table 2. Thus, it can be said that these models have classified all data inputs as abnormal which is not appropriate to select a model for decision support system. This work has also compared the performance of SVM model with other machine learning classifiers, that is, k-NN, ANN, and decision tree. In ANN decision support model, ReLU activation function was used with Adam optimizer, binary cross-entropy loss function and 100 epochs which produced more than 90% testing accuracy for all the three feature extraction techniques, namely, LBP, ALBP and RLBP. On the other hand, the performance of k-NN and Decision Tree has been lower than that of SVM and ANN classifier models. Although the best sensitivity and specificity scores were attained by LBP with SVM model, RLBP with SVM model, and ALBP with SVM model.

It can be observed that models such as LBP with SVM-AdaBoost fusion, LBP with SVM-XGBoost fusion, ALBP with SVM-AdaBoost fusion, ALBP with SVM-XGBoost fusion, RLBP with AdaBoost fusion, and RLBP with XGBoost fusion have not been able to correctly classify the normal

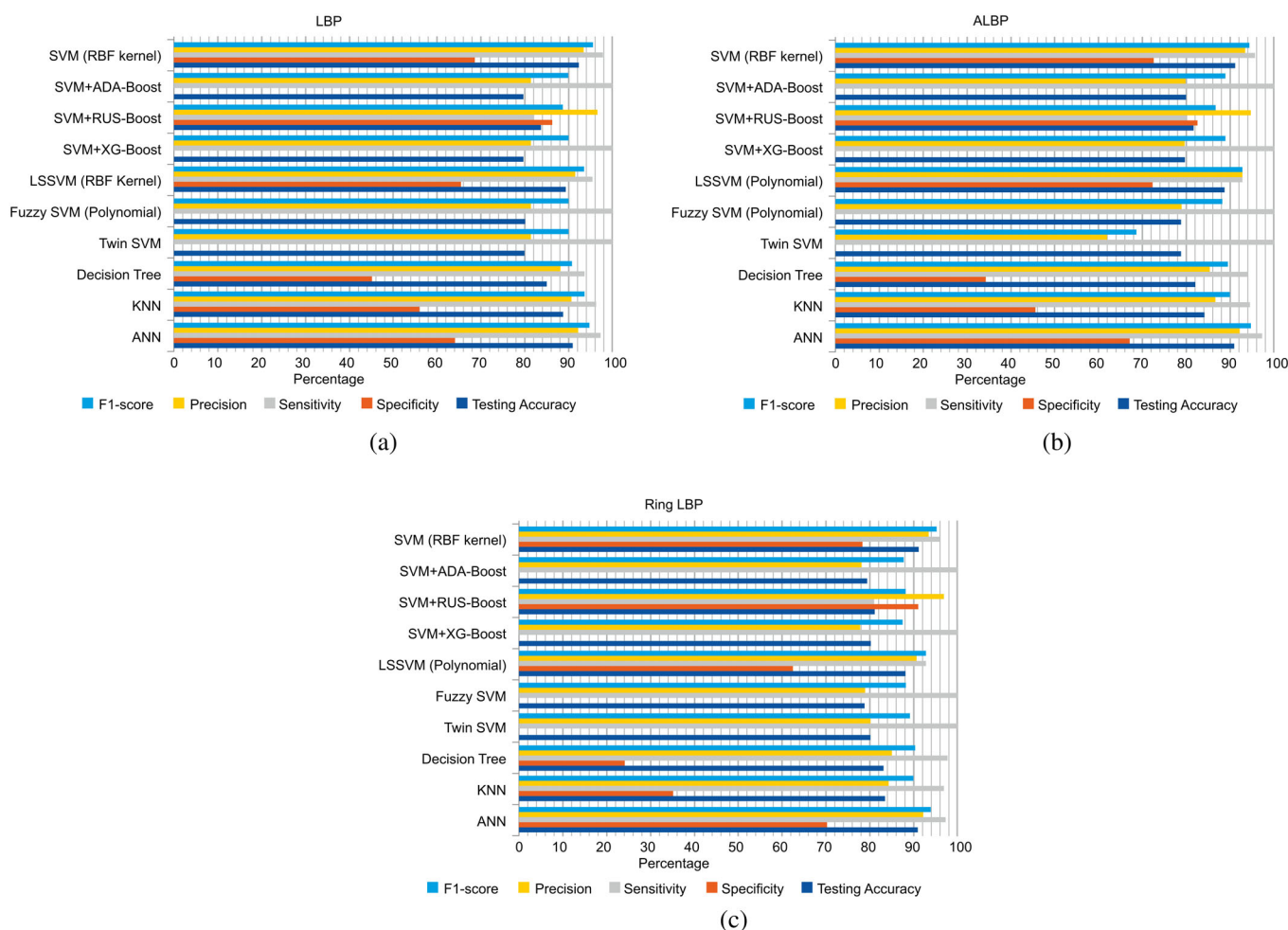


FIGURE 11 (a) Performance metrics of all models with linear binary pattern (LBP) features. (b) Performance metrics of all models with adaptive local binary pattern (ALBP) features. (c) Performance metrics of all models with ring LBP features.

class. In other words, the false-positive rate (FPR) has been quite high in the above-mentioned models for the heart sound classification of our dataset. RLBP with SVM and RUSBoost model, as reflected in Table 2, have recorded the best scores for mean precision relating to both abnormal and normal classes.

F1-score is a very useful metric in the case of imbalanced class distribution datasets. It balances the results of two metrics, precision, and sensitivity (also known as recall). In other words, it balances the metric for positive or negative samples which implies that it is a better metric to evaluate the performance of a machine learning model when the false-positive and false-negative values are crucial for a real-life problem. LBP with the SVM model recorded the highest F1-score of 95.27%. Similarly, RLBP with the SVM model also appeared with a good mean F1-score of 95.20% for both abnormal and normal classes. Table 2 reflects that the highest scores appeared in terms of sensitivity, specificity, testing accuracy, precision, and F1-score against each combination of specific classification models and type of features extracted. Further, the models where boosting algorithms were applied in fusion with SVM for classification purposes did not show better results than the models where only the SVM algorithm was used for classification. It has been due to the reason that when a classifier (mostly DT) is fused with boosting algorithm, it acts as a weak classifier; and boosting algorithm enhances that classifier's performance, but in this case SVM has not acted as a weak classifier when fused with boosting algorithms. Hence, contradiction exists with respect to the performance of both the algorithms (Wickramaratna et al., 2001). Further, all the three features, namely, LBP, RLBP, and ALBP performed extraordinarily with more than 90% testing accuracy and F1-score.

Receiver operator characteristic (ROC) curves are also used in binary decision problems for depicting the performance of machine learning algorithms. Figure 12 displays the ROC curves of the models that extracted LBP, RLBP, and ALBP features where the SVM algorithm was used for classification.

ROC curves are the probability curves plotted for true positive rate (TPR) against FPR at different threshold values. Area under curve (AUC) predicts the summary results of the plotted ROC curve. More the AUC value closer to 1, better would be the model prediction. The ROC-AUC

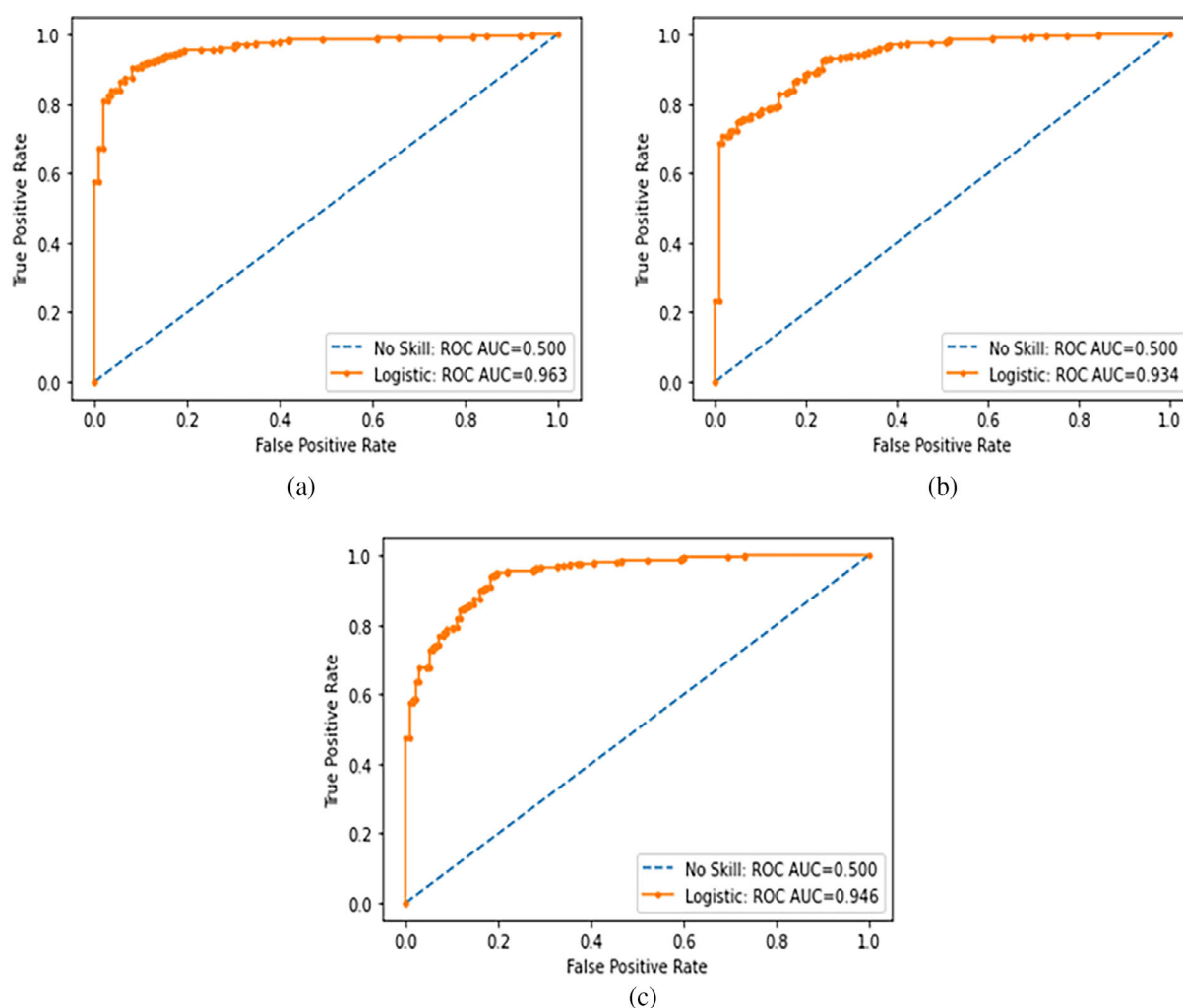


FIGURE 12 Receiver operator characteristic (ROC) curve for (a) linear binary pattern (LBP), (b) ring local binary pattern (RLBP), (c) adaptive local binary pattern (ALBP).

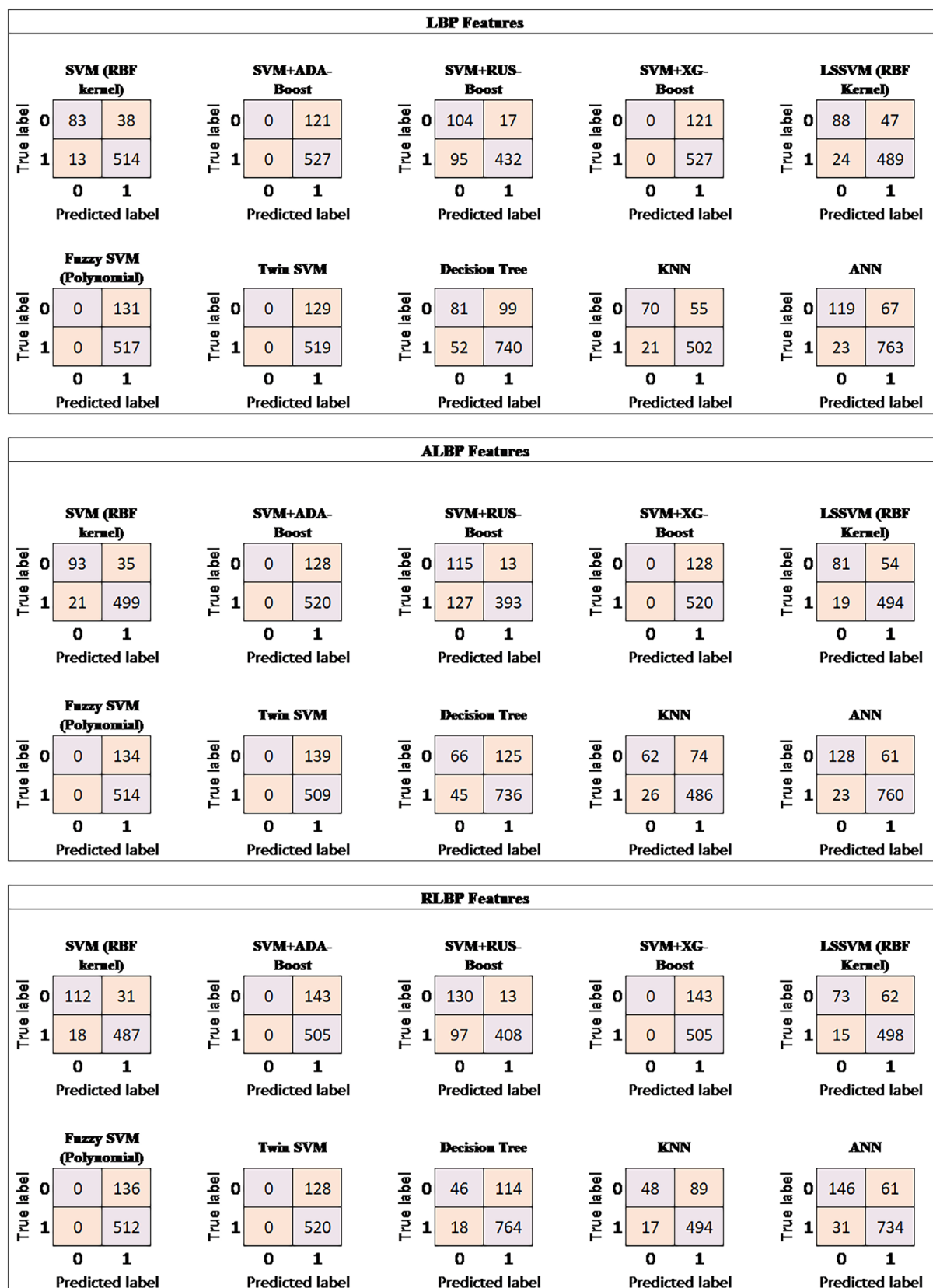


FIGURE 13 Confusion matrices.

value of the model, where LBP features were extracted and SVM used for classification, was comparatively higher than RLBP and ALBP features. The results of all the three feature-set models are sure to contribute significantly to the field under study. Figure 13 displays the confusion matrix of all the 30 models implemented in the proposed work for having a comparative analysis of various classifiers and feature extraction methods to know about the heart health status decision support system.

6 | CONCLUSION AND SCOPE FOR FURTHER RESEARCH

The unsegmented PCG signals were obtained from a reference dataset, PhysioNet 2016. The visual representation of the input audio signal was accomplished through the fusion of chromagram and spectrogram image extraction. The textural features of both chromagram and spectrogram images were then extracted using LBP and its variants ALBP and RLBP. The experimental results indicated that extracting LBP features from a chromagram-spectrogram fusion significantly improved the F1-score. The proposed model outperformed all other similar models mentioned in the literature with 92.12% accuracy and 95.27% F1-score. A comparative analysis of various machine learning algorithms used as classifiers indicated that the performance of traditional SVM was the best of all the classifiers. The optimization techniques such as various genetic algorithms or advance gradient descent techniques, and so forth can be incorporated for the purpose of future research. Several other variants of textural features and their fusion can also be investigated to further improve the results of heart sound classification.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in The PhysioNet/Computing in Cardiology Challenge 2016 at <https://physionet.org/content/challenge-2016/1.0.0/>.

ORCID

Kriti Taneja  <https://orcid.org/0000-0001-6970-416X>

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AUTHOR BIOGRAPHIES

Kriti Taneja is currently working as Research Scholar in Computer Science & Engineering Department at Thapar Institute of Engineering & Technology, Patiala, Punjab, India. Taneja received her M. Tech in Computer Applications degree from Thapar Institute of Engineering & Technology, Patiala, India in 2015. Her current interests focus on Machine learning and deep learning in healthcare.

Vinay Arora is working as Associate Professor in Computer Science & Engineering Department at Thapar Institute of Engineering & Technology, Patiala, Punjab, India. Arora was appointed in 2009, to Thapar Institute of Engineering & Technology, Patiala. Arora did his masters from Kurukshetra University, Kurukshetra, Haryana, India in 2007 and got campus placement in Newgen Software Technologies, New Delhi. He did his PhD in the domain of Model based software engineering and explored UML diagrams to full extent by creating one repository for UML activity diagrams. After PhD, he has shifted his research interests towards the application of machine/deep learning in Bio-medical signal processing and computational biology. He is also working with Environment & Energy department and Civil Engineering department for deploying machine/deep learning models to their specific problem domains.

Karun Verma was born in August 1980. He did his Bachelor's degree in Computer Science and Engineering from Punjab Technical University, Jalandhar India and Master's in Computer Science from Panjab University, Chandigarh, India. He received his Ph. D. from Thapar Institute of Engineering and Technology, Patiala and is working as an Associate Professor in the Department of Computer Science, Thapar Institute of Engineering and Technology, Patiala. He is currently pursuing research in the area of Data Science, statistical and machine learning based problems for recognition, classification, and information retrieval. He has more than twenty years of teaching experience. He had developed a complete working project on handwritten recognition for Indian regional language (Punjabi). He had also contributed to development of prosodically guided phonetic engine for searching speech databases for Indian Languages. Currently, he is working on projects building AI applications for precision agriculture. He has published various research papers in reputed national journals and conference proceedings. He has conducted various courses on Microsoft, Oracle, and Java technologies for students.

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